# Run report: target-free 4-sum system against the chained m = 3 target system (lead L1)

**Study** `research/target_free_relations_20261018/`, pre-registered in `PREREGISTRATION.md`. **Run** run-20261018T1645Z (2026-10-18 16:45 UTC; queue log 2026-10-18T16:45:52Z to 2026-10-18T18:27:36Z). **Commit** `aa7d718f0`. **Class:** stage diagnostic, with no S, no rho ratio and no speed claim.

## What ran

- **Command.** `run.sh`: per cell $(n, \ell) \in \{(16,4), (18,4), (20,5), (24,6)\}$, basis `gp` (one draw: the progression base is deterministic) and `random` (four draws), twists `[0,0,0,0]` and `[0,1,2,3]`, one process per (cell, basis, twists): `prlimit --as=12000000000 timeout 7200 dreg_target_free --cells n:ℓ --basis B --twists T --draws K --seed 20261018 --budget-secs 1200 --target --no-count`, with `F4_F2_MAX_ROWS=F4_F2_MAX_COLS=50000000`, `RAYON_NUM_THREADS=2` and the default 1 GiB F4 step-matrix cap.
- **Arms.** `target_free`: `build_target_free_system(basis, b = 1, twists)`, $4\ell + n$ unknowns, $2n$ equations. `target_m3`: `build_decomposition_system(basis, x_R, b = 1, m = 3)` on the same basis with a fresh random `x_R` per draw, $3\ell + n$ unknowns, $2n$ equations. Both solved by `pq_f4_f2::groebner_basis_f4` with the field equations.
- **Measure.** F4 solving degree (highest step degree that added a basis element) and degree reached. A record with `oversize` or `timed_out` is unmeasured; its `solving_degree` is a lower bound only.
- **Host.** The 4-core, 15 GB container, which is not isolated (`AGENTS.md` §10). Wall times are practicality notes only.
- **Scorer.** `dreg_score target_free runs/cell-*.jsonl`, output in `score-output.txt`.

## Completion

4 of 4 `runs/cell-*.jsonl` records non-empty; 4 processes logged an exit; the logged statuses are not the processes' (the script read `$?` after a command substitution, so every logged status is that of `date`; see the hand note at the end of `runs/queue.log`, fixed in `run.sh` afterwards). Every record here is non-empty and its `.log` is empty. A fifth process, `cell-18-4.gp.tw0000`, was killed within seconds of its start when the queue was stopped at the registered stop condition (target-free arm unmeasured on every (16, 4) draw); its empty record was removed and the stop is noted in the queue log. Target-free arm: 10 draws, 0 finished, 10 oversize, 0 timed out. Target arm: 10 draws, 10 finished, 0 oversize, 0 timed out. Longest single F4 run 168 s; largest recorded step matrix 0.35 GiB. Peak RSS was not recorded by the harness.

## Measured values

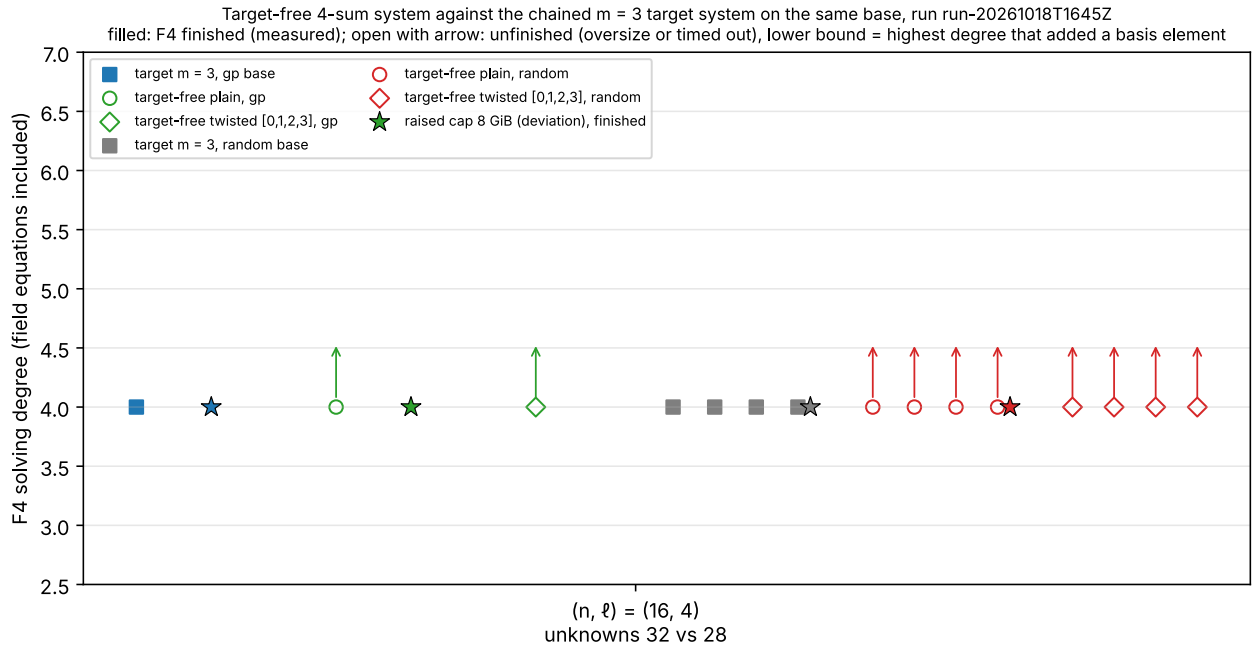



Figure 1: Recorded values from `runs/cell-*.jsonl`, run run-20261018T1645Z. F4 solving degree per draw and arm; filled markers are finished runs, open markers with an arrow are unfinished runs drawn at their lower bound. Source: `plot.py` (layout only).

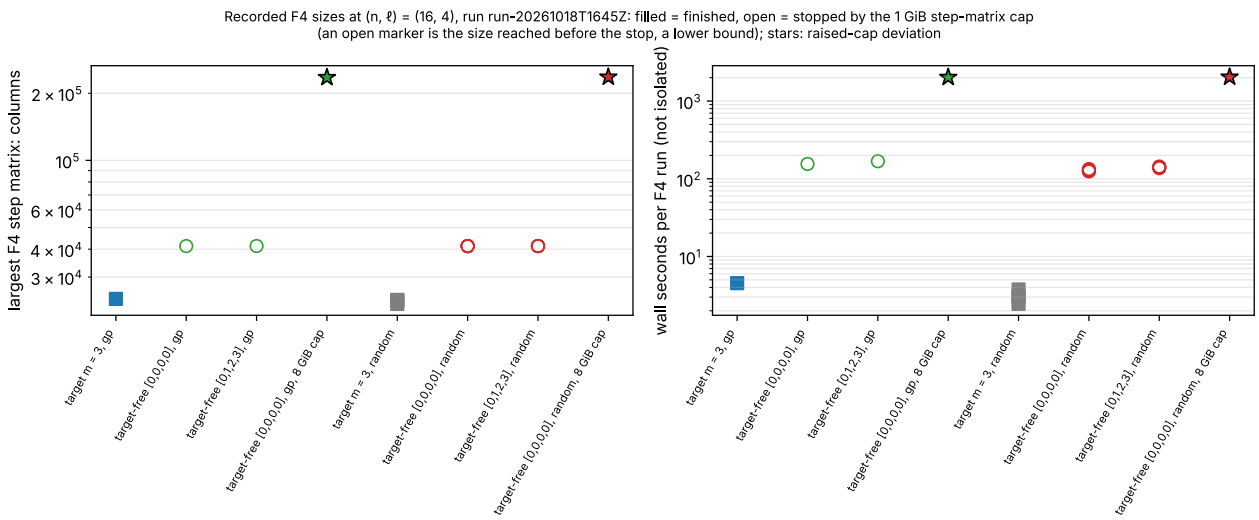



Figure 2: Recorded `matrix_cols_max` and `secs` per F4 run at (16, 4), run run-20261018T1645Z, log scale. Open markers stopped at the 1 GiB step-matrix cap, so their values are what was reached before the stop. Wall seconds are practicality notes on a non-isolated host. Source: `plot.py`.

### Per (cell, basis, twists) readings

| (n, ℓ) | basis | twists | target-free measured / draws | target-free solving (measured; bounds ≥) | target m = 3 measured / draws | target solving (measured) | unsat. target draws | registered reading | single-arm reading (fewer than 3 draws) |
|---|---|---|---|---|---|---|---|---|---|
| (16, 4) | gp | [0,0,0,0] | 0 / 1 | bounds ≥ 4 | 1 / 1 | 4 | 1 | unmeasured | not lower (bounded: every target-free draw ≥ target median) |
| (16, 4) | gp | [0,1,2,3] | 0 / 1 | bounds ≥ 4 | 1 / 1 | 4 | 1 | unmeasured | not lower (bounded: every target-free draw ≥ target median) |
| (16, 4) | random | [0,0,0,0] | 0 / 4 | bounds ≥ 4, 4, 4, 4 | 4 / 4 | 4, 4, 4, 4 | 4 | unmeasured | not lower (bounded: every target-free draw ≥ target median) |
| (16, 4) | random | [0,1,2,3] | 0 / 4 | bounds ≥ 4, 4, 4, 4 | 4 / 4 | 4, 4, 4, 4 | 4 | unmeasured | not lower (bounded: every target-free draw ≥ target median) |

### Registered readings (from `dreg_score target_free`)

```
Registered verdict: killed on the bounded reading (no cell reads lower for gp on any draw, measured or bounded); undecided by the registered rule (fewer than 3 measured draws per arm). gp cells reading lower: []; random cells reading lower: []; gp cells with a registered reading: 0 of 1; gp cells bounded not-lower on every twist: 1 of 1.
P1 (gp plain reads higher or equal at every measured cell): unmeasured — (16,4)[0000]: unmeasured, bounded not lower
P2 (gp twisted reads equal at every measured cell): unmeasured — (16,4)[0123]: unmeasured, bounded not lower
P3 (random never reads lower): unmeasured — (16,4)[0000]: unmeasured, bounded not lower; (16,4)[0123]: unmeasured, bounded not lower
P4 (target arm F4 solving degree = 4 on unsatisfiable draws at (16,4), (18,4)): holds — 10 of 10 unsatisfiable measured target draws read 4
```

## Deviation: raised-cap pass on (16, 4)

Not registered (see `RESULTS.md`, "Deviations"). `run_highcap.sh`: the same arms at (16, 4), one draw each (gp plain, gp twisted, random plain, seed 20261018, the first draw of `runs/`), with `F4_F2_MAX_MATRIX_WORDS = 2^30` (8 GiB step matrices), a 3,600 s budget and the same 12 GB address-space limit. Records: `runs_highcap/`; scored separately, never pooled with `runs/` (`score-output-highcap.txt`). Stars in the figure.

| (n, ℓ) | basis | twists | target-free measured / draws | target-free solving (measured; bounds ≥) | target m = 3 measured / draws | target solving (measured) | unsat. target draws | registered reading | single-arm reading (fewer than 3 draws) |
|---|---|---|---|---|---|---|---|---|---|
| (16, 4) | gp | [0,0,0,0] | 1 / 1 | 4 | 1 / 1 | 4 | 1 | unmeasured | equal (1 vs 1 measured draws) |
| (16, 4) | random | [0,0,0,0] | 1 / 1 | 4 | 1 / 1 | 4 | 1 | unmeasured | equal (1 vs 1 measured draws) |

Queue log (`runs_highcap/queue.log`):

```
2026-10-18T17:16:02Z start research/target_free_relations_20261018/runs_highcap/cell-16-4.gp.tw0000
2026-10-18T17:49:49Z exit 0 research/target_free_relations_20261018/runs_highcap/cell-16-4.gp.tw0000
2026-10-18T17:49:49Z start research/target_free_relations_20261018/runs_highcap/cell-16-4.gp.tw0123
2026-10-18T17:53:13Z exit 0 research/target_free_relations_20261018/runs_highcap/cell-16-4.gp.tw0123
2026-10-18T17:53:13Z start research/target_free_relations_20261018/runs_highcap/
cell-16-4.random.tw0000
2026-10-18T18:27:15Z exit 0 research/target_free_relations_20261018/runs_highcap/
cell-16-4.random.tw0000
2026-10-18T18:27:36Z note written by hand: the 'exit' statuses above are date's, not the runs'
(run_highcap.sh read $? after a command substitution; fixed afterwards). cell-16-4.gp.tw0123 aborted
on allocation under the 12 GB address-space limit (its .log; empty record); the other two finished.
```

## Observations (no interpretation)

See `RESULTS.md` for the reading against the registered verdict rules and the predictions; this report records what ran and what was measured.